

Before class: what is in the IV bag?

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About 15 minutes. Read the four short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. A periodic table and the answer key are at the back.

1 · Read first (5 min)

Solute and solvent

A **solution** is a mixture so even that you cannot see its parts. The substance present in the largest amount is the **solvent** (usually water); what is dissolved in it is the **solute**. Normal saline is NaCl (solute) in water (solvent).

Like dissolves like (recap)

From Ionic & Covalent II: water is a **polar** molecule that can hydrogen bond. Polar and ionic substances dissolve in water; **nonpolar** substances such as oils dissolve in nonpolar solvents.

How much is dissolved?

The **concentration** tells how much solute is in a given amount of solution. Labels use percent (0.9% NaCl), mg per mL (paracetamol syrup 120 mg/5 mL) or **molarity**: moles of solute per litre of solution. You already know how to turn grams into moles (Chemical Reactions I).

Water follows the particles

Cell membranes let water through but hold back many dissolved particles. Water moves towards the side with **more** dissolved particles. This is **osmosis**, and it is why IV fluids must match the particle concentration of blood.

2 · Readiness check

Ten quick questions on what you need for today.

Q1 SOL-L01

In a sugar solution made by dissolving sugar in water, the solvent is:

- A. sugar
- B. water
- C. the solution
- D. the container

Q2 SOL-L01

A mixture of salt and pepper is a solution.

True / False

Q3 SOL-L01

Recap: which substance is most likely to dissolve in water?

- A. cooking oil
- B. hexane, C_6H_{14}
- C. sodium chloride, NaCl
- D. candle wax

Q4 SOL-L01

Recap: water is a good solvent for polar substances mainly because it:

- A. is a liquid
- B. is polar and can form hydrogen bonds
- C. has a low boiling point
- D. contains no ions

Q5 SOL-L02

Recap: the molar mass of NaCl (Na = 22.99, Cl = 35.45).

Q6 SOL-L02

Recap: how many moles are in 11.7 g of NaCl (58.44 g/mol)?

Q7 SOL-L02

Convert 250 mL to litres.

Q8 SOL-L02

A saline bag is labelled 0.9% NaCl. This most likely means:

- A. 0.9 g NaCl in 100 mL of solution
- B. 0.9 g NaCl in 1 L of water
- C. 0.9 mol NaCl per litre
- D. 9 g NaCl in 100 mL

Q9 SOL-L03

When water is added to a solution, the amount (moles) of solute stays the same.

True / False

Q10 SOL-L05

Water moves across a cell membrane towards the side that has:

- A. fewer dissolved particles
- B. more dissolved particles
- C. more water molecules
- D. a higher temperature

3 • Think ahead

No wrong answers here. Make a guess, then check it after the lecture (Parts 5 and 6).

Q11 SOL-L05

Why would a nurse never infuse pure sterile water into a vein, when normal saline is given every day?

Q12 SOL-L04

Why do icy roads get salted in cold countries, and why does car coolant contain ethylene glycol?

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

Before class: what is in the IV bag?

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About 15 minutes. Read the four short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 SOL-LO1

In a sugar solution made by dissolving sugar in water, the solvent is:

- A. sugar
- B. water**
- C. the solution
- D. the container

The solvent is the major component; water is the solvent and sugar the solute.

Q2 SOL-LO1

A mixture of salt and pepper is a solution.

True / False → **False**

You can see the separate grains, so it is not homogeneous. Salt dissolved in water is a solution.

Q3 SOL-LO1

Recap: which substance is most likely to dissolve in water?

- A. cooking oil
- B. hexane, C_6H_{14}
- C. sodium chloride, NaCl**
- D. candle wax

Ionic NaCl dissolves in polar water; the others are nonpolar.

Q4 SOL-L01

Recap: water is a good solvent for polar substances mainly because it:

- A. is a liquid
- B. is polar and can form hydrogen bonds**
- C. has a low boiling point
- D. contains no ions

Similar intermolecular forces let water surround polar and ionic solutes.

Q5 SOL-L02

Recap: the molar mass of NaCl (Na = 22.99, Cl = 35.45).

58.44 g/mol

$22.99 + 35.45 = 58.44 \text{ g/mol}$.

Q6 SOL-L02

Recap: how many moles are in 11.7 g of NaCl (58.44 g/mol)?

0.2 mol

$11.7 \div 58.44 = 0.200 \text{ mol}$.

Q7 SOL-L02

Convert 250 mL to litres.

0.25 L

$250 \div 1,000 = 0.250 \text{ L}$.

Q8 SOL-L02

A saline bag is labelled 0.9% NaCl. This most likely means:

- A. 0.9 g NaCl in 100 mL of solution**
- B. 0.9 g NaCl in 1 L of water
- C. 0.9 mol NaCl per litre
- D. 9 g NaCl in 100 mL

For IV fluids the percent is mass/volume: grams per 100 mL of solution.

Q9 SOL-L03

When water is added to a solution, the amount (moles) of solute stays the same.

True / False → **True**

Only the volume changes, so the concentration falls. This is the idea behind dilution.

Q10 SOL-L05

Water moves across a cell membrane towards the side that has:

- A. fewer dissolved particles
- B. more dissolved particles**
- C. more water molecules
- D. a higher temperature

Water moves into the more concentrated solution: osmosis.

3 · Think ahead

Q11 SOL-L05

Why would a nurse never infuse pure sterile water into a vein, when normal saline is given every day?

Pure water has far fewer dissolved particles than blood (it is **hypotonic**). Water would move into red blood cells by osmosis until they burst (**haemolysis**). Normal saline has the same osmolarity as blood (**isotonic**). See Part 6 of the slides.

Q12 SOL-L04

Why do icy roads get salted in cold countries, and why does car coolant contain ethylene glycol?

Dissolved particles **lower the freezing point** and **raise the boiling point** of water (colligative properties). Salt water freezes below 0 °C; coolant boils above 100 °C. See Part 5.

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